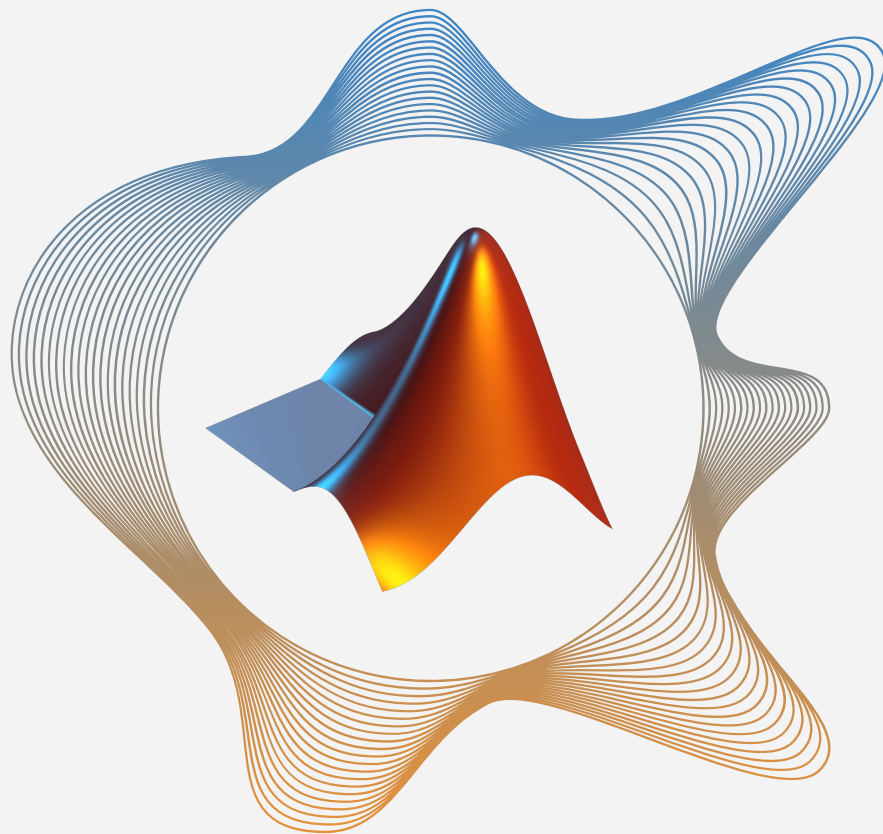




MathWorks Denoising Speech

By: Ella, Rianna, Jenny, Andie,
Sowndaryan, Lynn

Challenge Advisor: Neha Sardesai



Meet Our Team



Andrea Cabochan
Tufts University



Ella Hou
Tufts University



Jenny Li
Boston University



Qien Lin
Brown University



Rianna James
Boston University



**Sowndaryan
Jayaprakash**
UMass Lowell

Our AI studio Coach & Challenge Advisor

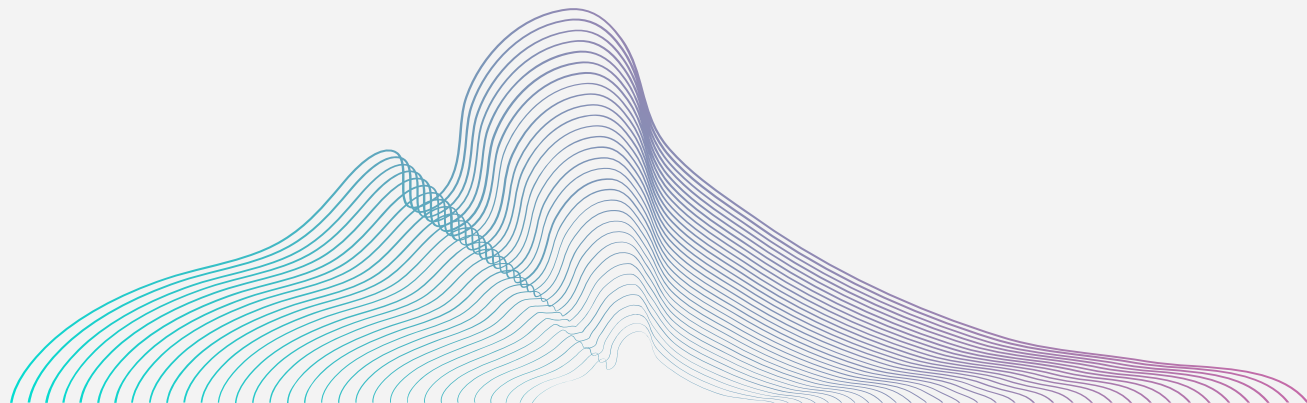


**Rawisara (Mimi)
Lohanimitt**



**Neha
Sardesai**

Overview





Introduction

Goal: Build a deep learning neural network capable of **suppressing background noise** in speech signals

Additional Objectives:

*stretch goal



**Speech Quality
Preservation**



**Robust Noise
Environments**



**Real-Time
Processing***



**Hearing Aid
Integration***



Project Impact

Suppressing background noise has two main use cases:

Hearing Aids

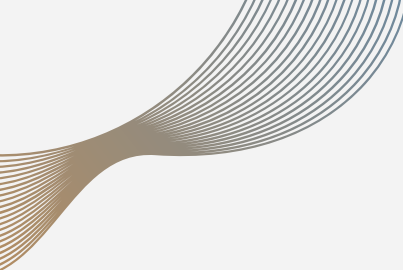
Enhances speech clarity for people with hearing impairments by reducing noise

Virtual Meetings

Reduces background distractions for a more natural, focused listening experience

Our project is useful to these MathWorks teams: Customer Support, Application Engineering, and Audio Processing





Vocabulary

01

LSTMs (Long Short-Term Memory)

A Recurrent Neural Network that preserves long-term dependencies, useful for language and speech tasks

02

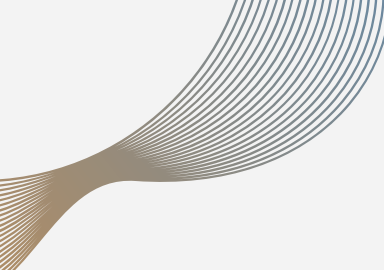
Convolutional Neural Network (CNN)

A neural network using convolutional filters to learn spatial features, useful for short-term pattern recognition

03

Fourier Transform

A function that converts audio from the time domain to the frequency domain.



Vocabulary

04

Short-Term Fourier Transform (STFT)

Chopping the audio into small slices and applying the Fourier Transform onto each slice

05

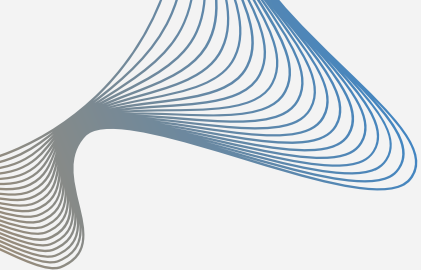
Signal-to-Noise Ratio (SNR)

A logarithmic measure (dB) of signal vs. noise power. Higher SNR is cleaner.

06

Short-Time Objective Intelligibility (STOI)

A metric of speech intelligibility, ranging from 0 to 1. Higher STOI is more understandable



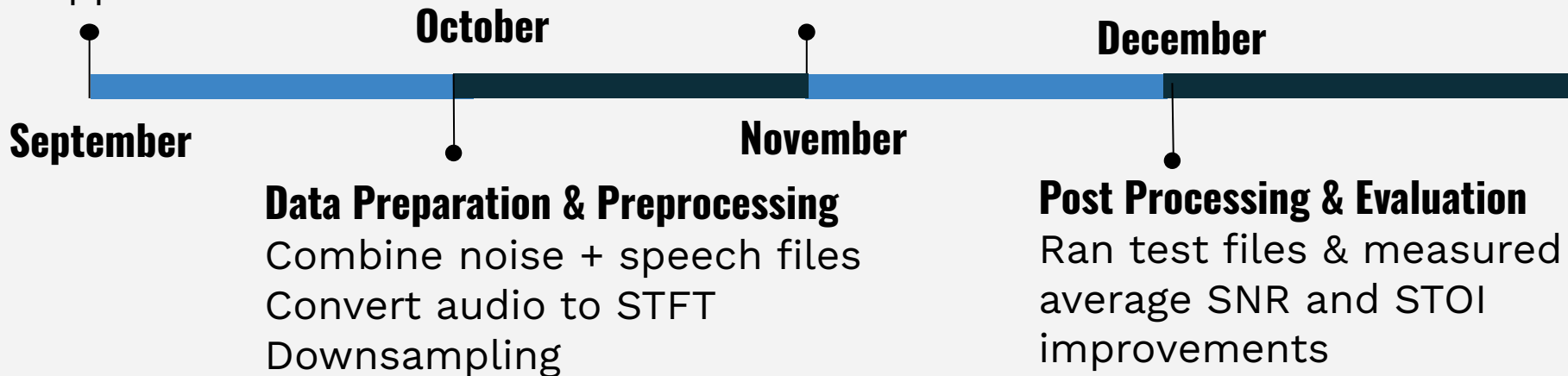
Our Approach

Literature review

Researched deep learning approaches for noise suppression

Model Building

Started building the CNN & LSTM models

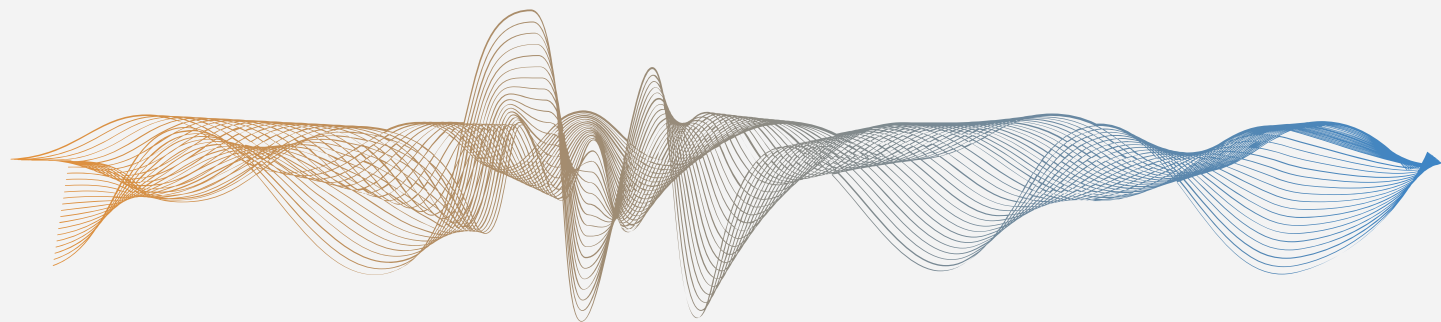




DATA

Microsoft DNS Challenge

Led by Microsoft and ICAAP*, inviting researchers worldwide to develop AI models for denoising using a standardized dataset

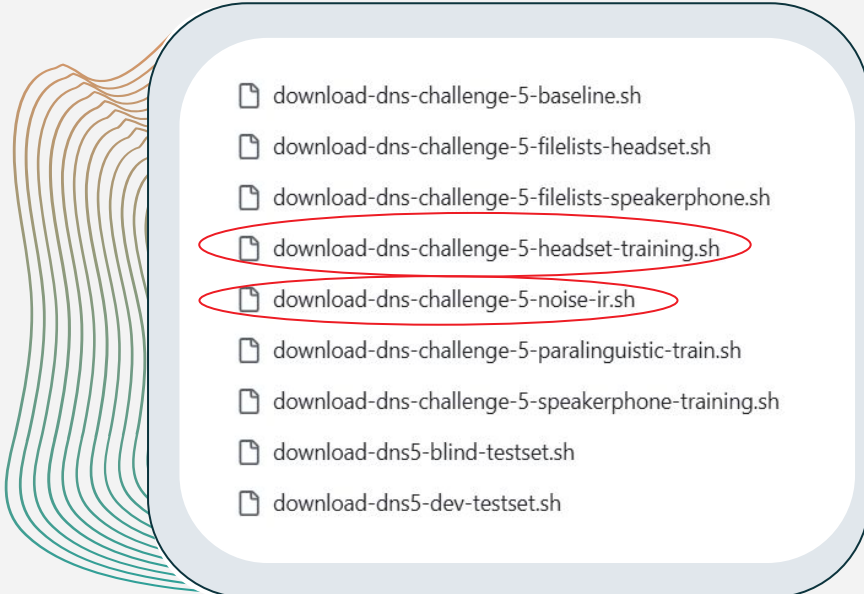


DNS5 Dataset Structure

Contains audio data available through [Github Repo](#):

- 1 TB of clean speech from headsets
 - Includes multiple languages, accents, even singing
- 39 GB noisy data from different environments
 - Examples include static, traffic, music, animals

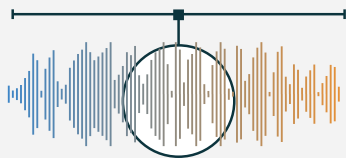
Raw format: .WAV files with 48 kHz sampling rate



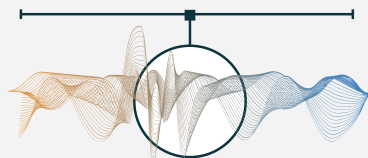
A list of shell scripts for downloading DNS5 dataset components, enclosed in a light blue rounded rectangle. To the left of the rectangle are several wavy lines in orange and green. Two items in the list are circled in red: 'download-dns-challenge-5-headset-training.sh' and 'download-dns-challenge-5-noise-ir.sh'.

- download-dns-challenge-5-baseline.sh
- download-dns-challenge-5-filelists-headset.sh
- download-dns-challenge-5-filelists-speakerphone.sh
- download-dns-challenge-5-headset-training.sh
- download-dns-challenge-5-noise-ir.sh
- download-dns-challenge-5-paralinguistic-train.sh
- download-dns-challenge-5-speakerphone-training.sh
- download-dns5-blind-testset.sh
- download-dns5-dev-testset.sh

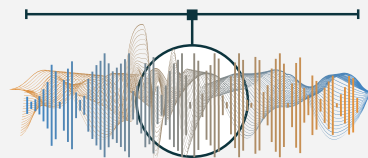
Data Flow - Noise Suppression



**Clean Speech
(Actual)**



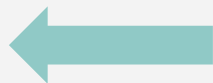
Noise Clip



**Noisy Sample
(Model Input)**



**Clean Speech
(Predicted)**



Apply inverse
STFT
Multiply IRM by
mixed input



**Ideal Ratio Mask
(Model Output)**

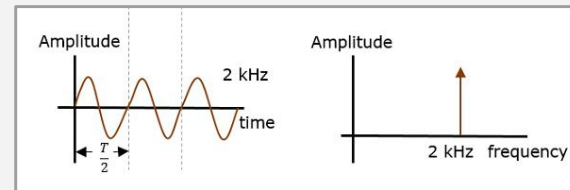


Apply STFT and
feed into model



PREPROCESSING

Preprocessing Steps



1. Combine Speech & Noise:

- ~~$10 \log_{10} \frac{P_{\text{noise}}}{P_{\text{speech}}}$~~ *
- Match lengths

-11.82 dB to 12.56 dB

2. Downsample:

- **48 kHz $\rightarrow$ 16 kHz**
- Reduce storage + computation time

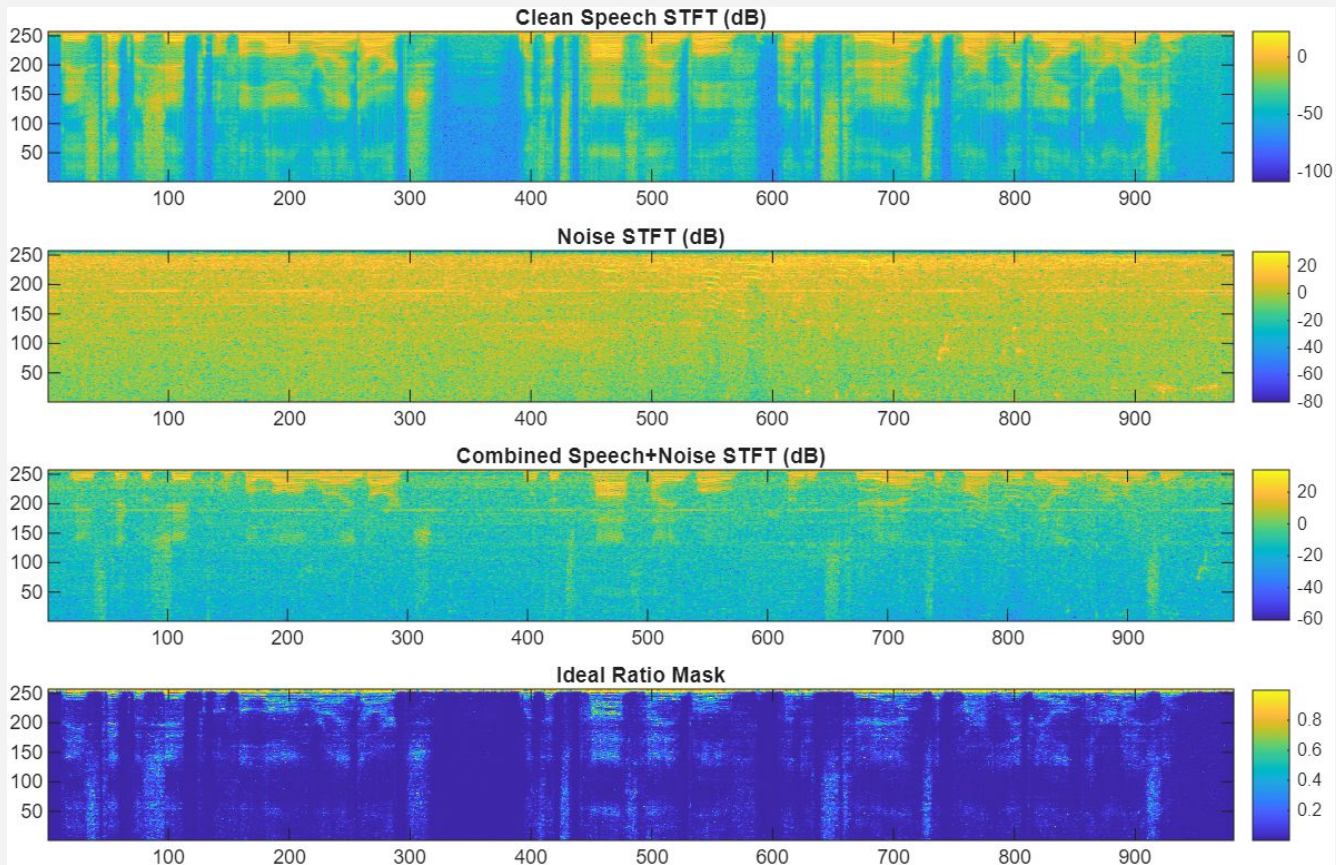
3. STFT:

- Slice into frames
- Time $\rightarrow$ Frequency
- **512-sample** Hamming window, **75% overlap**, **257 frequency bins**

4. Compute Ideal Ratio Mask (IRM)

- $$\frac{|\text{clean speech}|}{|\text{clean speech}| + |\text{noise}|}$$
- Outputs $\in [0, 1]$

STFT Visualization





MODEL BUILDING

<u>Paper</u>	<u>Key Insight Applied</u>
Cross-Lingual Multi-Granularity Framework	Deep LSTM stacks (6 layers) + dropout (0.3) achieve robust temporal modeling
Hindi Speech Separation	CNN→LSTM hybrid balances noise suppression with feature preservation; 1-3 layers optimal
Audio-Visual Speech Enhancement	CNN preprocessing reduces LSTM input complexity → faster convergence & lower latency

- **Parameter Selection Rationale:**

- **Layers:** 3 LSTM layers (literature: 1-3 optimal for speech)
- **Hidden units:** 128-256 range (balance capacity vs. efficiency)

Model Building

Convolutional Neural Network

- Excels at processing grid data (images, time series, **spectrograms**)
- Used to process time-frequency representation

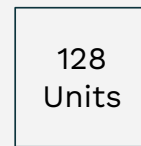
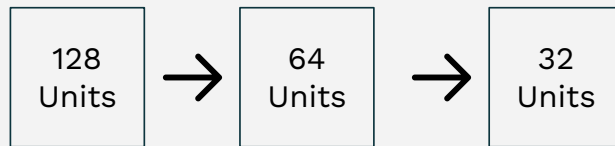
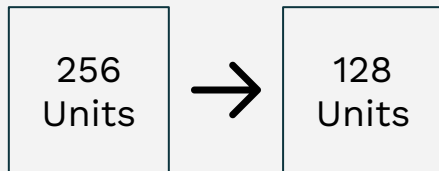
Long Short Term Memory

- Used to map temporal dependencies between current frame and previous frames, mapping rhythm and flow



Pipeline

Early LSTMs → **CNN Block** → **Final LSTM**



Model Architecture

- **Model Architecture:** CNN + LSTM
 - **CNN:** extract local spectral patterns -> reduce background noise
 - **LSTM:** captures long-term temporal dependencies → models speech rhythm and flow
 - **Combined Strength:** CNN handles frequency features and LSTM manages time dependencies
- **Our Architecture Design:**
 - **Pipeline:** LSTM → LSTM → CNN Stack → LSTM → FC
 - **Early LSTMs** (256→128 units): Learn temporal context from raw features
 - **CNN Block** (3 conv layers: 128→64→32 filters): Extract hierarchical frequency patterns
 - **Final LSTM** (128 units): Temporal integration & sequence summarization/preservation (spectrogram output)





MODEL TRAINING

Model Training By The Numbers

> 29,000 speech
files (VCTK)

> 8000 noise files
(fullband 000)

English-
speakers
reading full
sentences

6000 combined files
1-12 seconds long

4000 training
(66.67%)

1000 validation
(16.67%)

1000 testing
(16.67%)

Model Training Details

Training Configuration

- Framework: **MATLAB trainNetwork** (sequence regression)
- Optimizer: **Adam**
- Max Epochs: **50**
- Mini-Batch Size: **8** (for variable-length version)
- Initial Learning Rate: **0.001**
- Gradient Clipping: **GradientThreshold = 1**

Loss Function

Training loss:

- MATLAB → **Mean Squared Error (MSE)**
- Sequence-to-sequence regression on **257-bin spectra / ratio masks** over time



Model Training Details

Framework: **MATLAB** **trainNetwork** (sequence regression)

Optimizer

Adam

Max Epochs

50

Batch Size

8

Learning Rate

0.001

Gradient Clipping

Threshold = 1

Loss Function

MSE



POSTPROCESSING

Post Processing

Goal: Convert the model's predicted magnitudes into real, time domain audios (.wav)

1. **Load:** Model-predicted magnitudes and noisy STFT (for phase reference)
2. **Align:** Match time frames between predicted and noisy spectra
3. **Rebuild STFT:** Combine predicted magnitudes + noisy phase, mirror frequencies
4. **Inverse STFT:** Convert back to time-domain waveform
5. **Normalize & Output:** Prevent clipping → visualize, play, or save audio

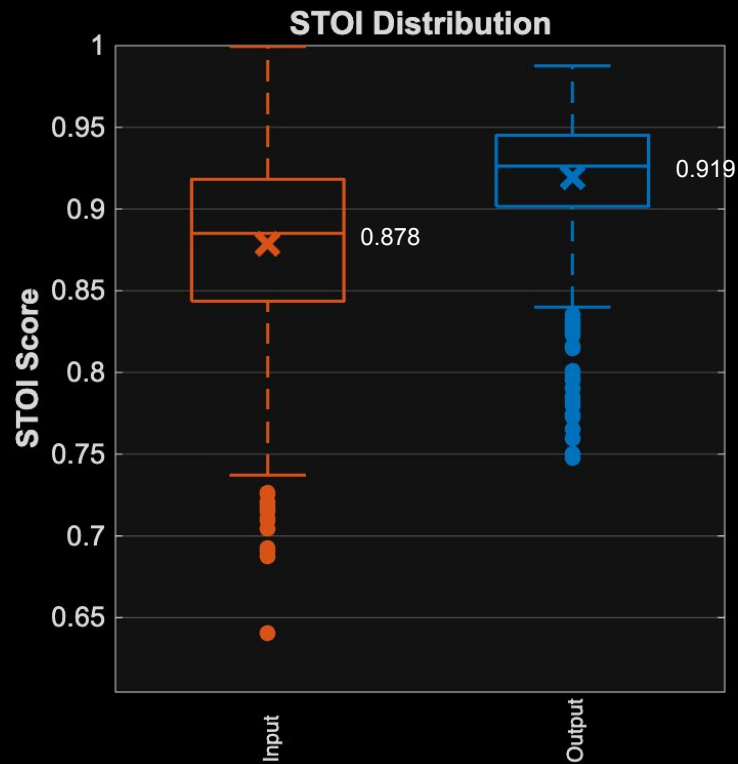
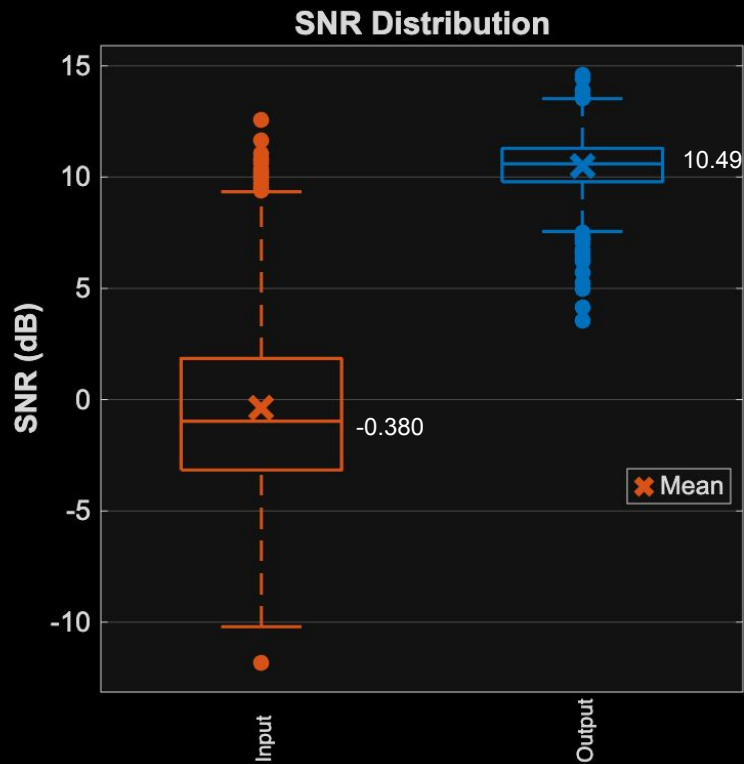




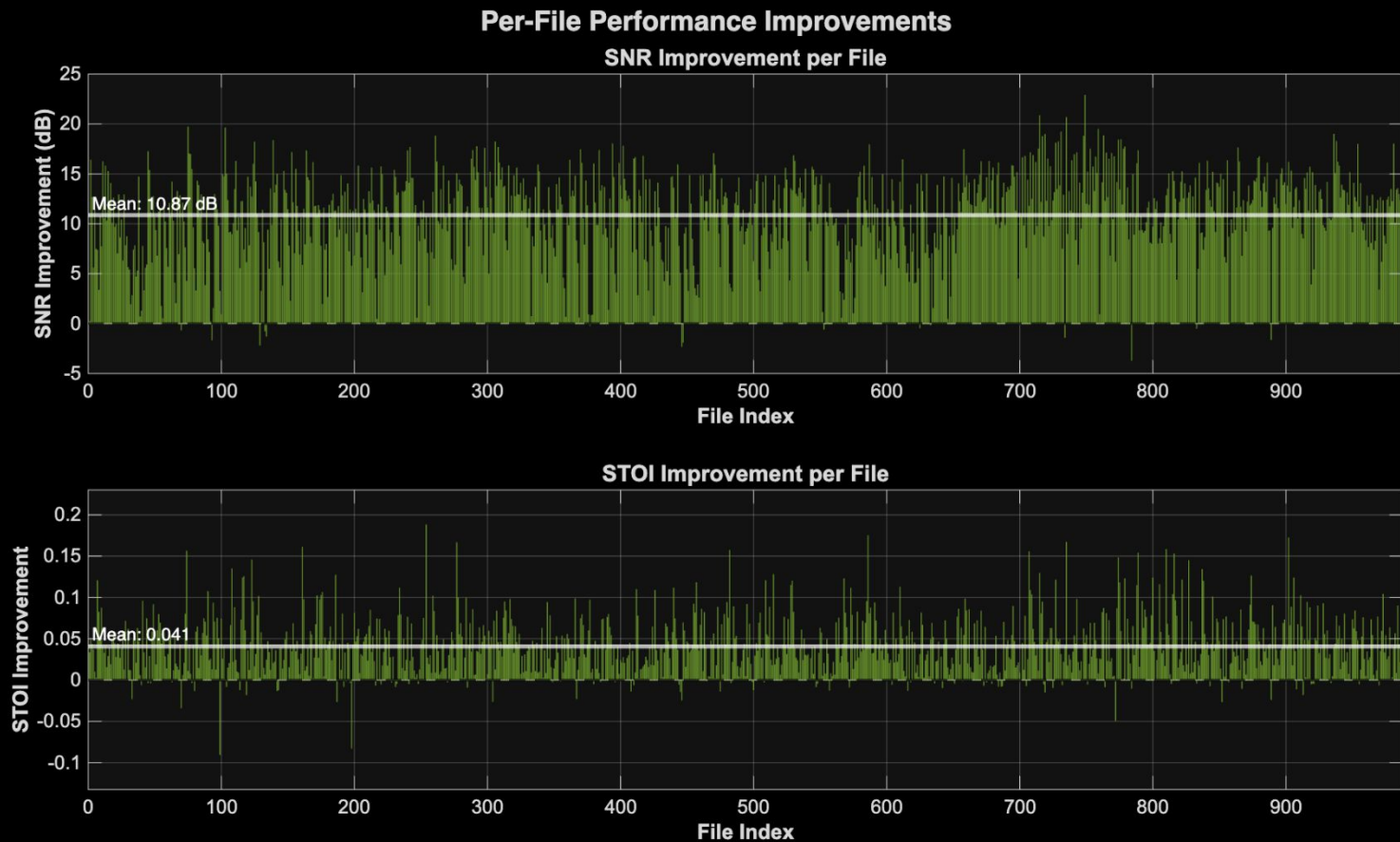
RESULTS

Overall Results

Performance Distribution Comparison

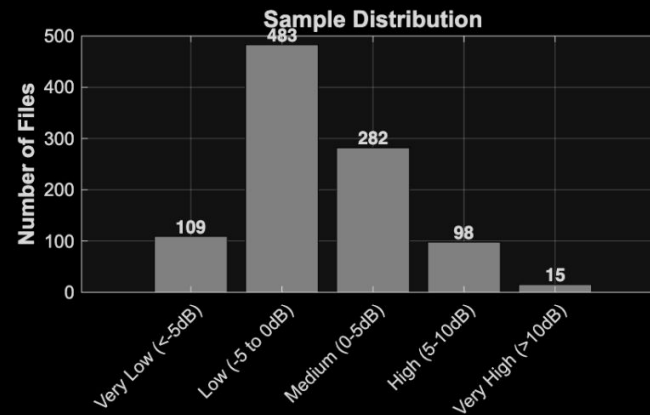
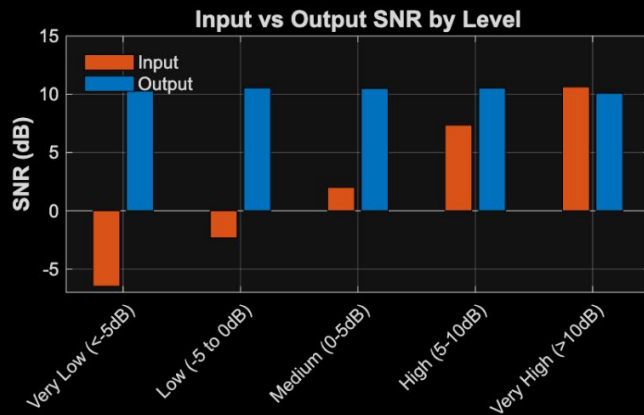
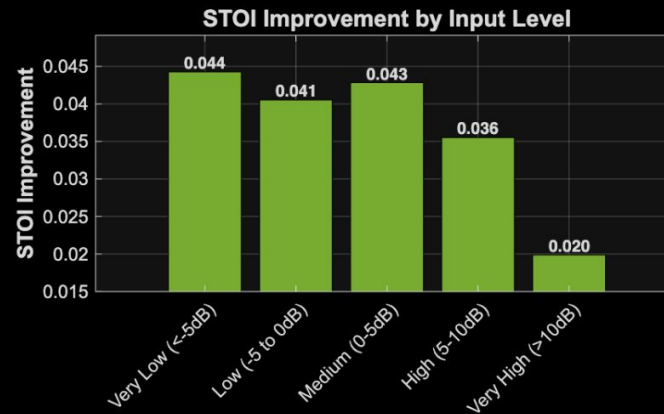
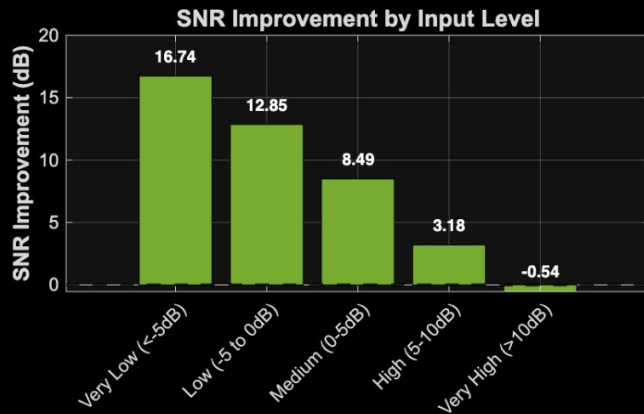


Overall Results



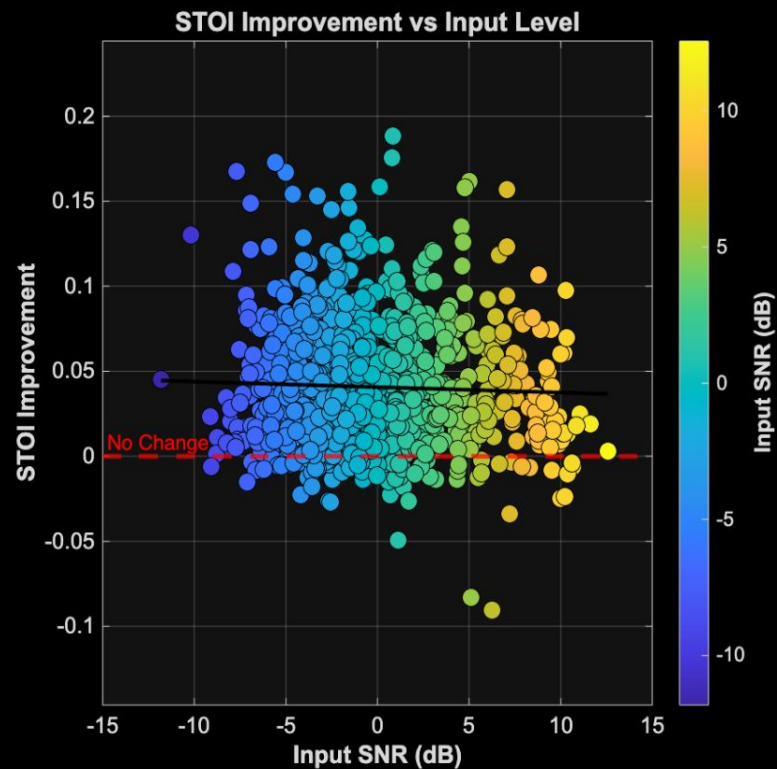
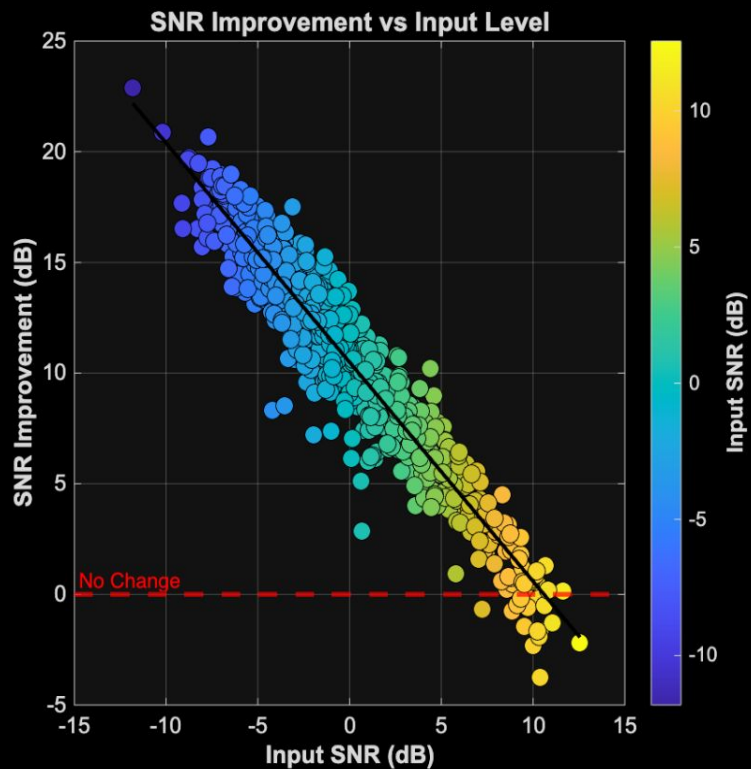
Stratified Results

Stratified Performance Analysis



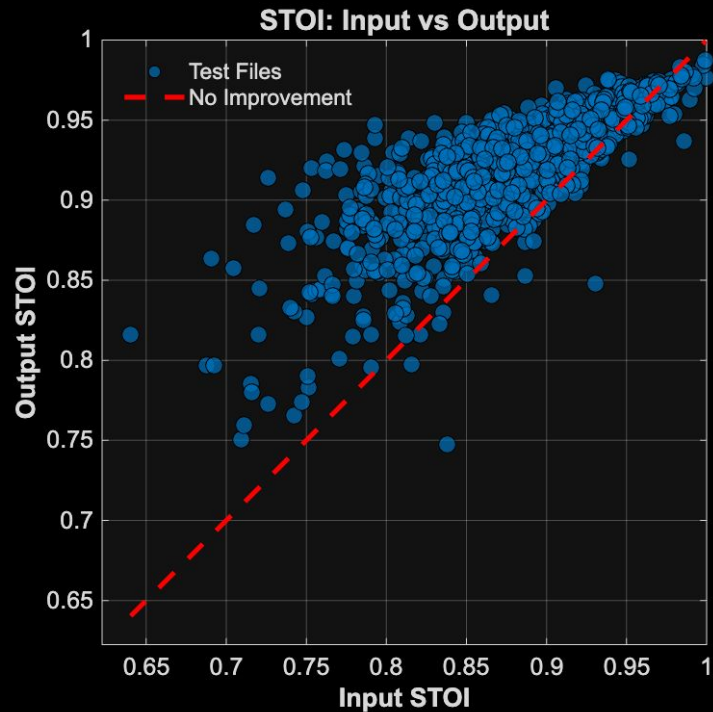
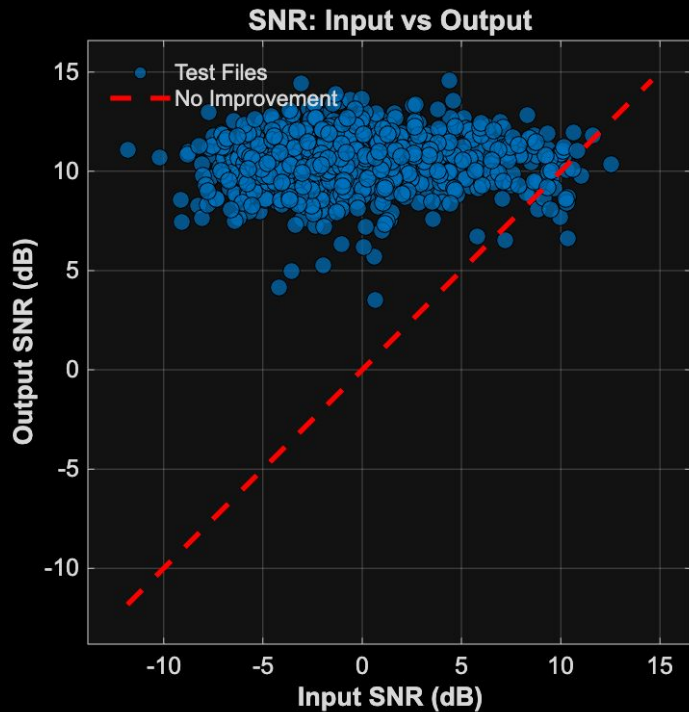
Stratified Results

Improvement Analysis by Input SNR



Stratified Results

Input vs Output Performance



Perceptual Analysis

Just Noticeable Difference (JND): Smallest amount of improvement a human listener is likely to perceive

- SNR: **3 dB** - found by McShefferty et. al
- STOI: no official finding, but we will use **0.05** as an estimate

Files showing improvement:

SNR: 972/987 (98.5%)

STOI: 893/987 (90.5%)

Files > JND:

SNR: 932/987 (94.4%)

STOI: 327/987 (33.1%)

Perceptual Analysis



(input)



(output)



(input)



(output)



(input)



(output)



(input)



(output)



(input)



(output)

----- Common side effect: decreased amplitude? -----

Key Findings



Strong Overall Performance

- Mean SNR improvement = 10.87 dB
- Mean STOI improvement = 0.041
- Statistically significant (< 0.0001) using t-tests



Inputs with low SNR inputs

- < -5 dB bin: +16.74 dB SNR, +0.044 STOI
- Equalizer effect: Output SNR comparable to outputs of high SNR inputs (~ 10 dB)



Key Findings



Inputs with high SNR

- > 10 dB bin: -0.54 dB SNR, $+0.020$ STOI
- Possible cause: under-representation of high SNR samples in training set



Perceptual Anomalies

- Decreased amplitude
- Possible cause: over-suppression affecting not only noise, but also speech





CHALLENGES

Challenges

Audio length mismatch after inverse STFT

Half-spectrum vs. full-spectrum reconstruction mismatches

Magnitude and mask matrix size inconsistencies

Negative SNR outputs from phase misalignment in evaluation

Solutions

We trimmed both clean and predicted signals to a shared minLen before evaluation.

We rebuilt the missing negative-frequency bins using conjugate symmetry to ensure a valid complex STFT.

Clipping and resizing ensured that element-wise multiplication (`mixtureMag.* ratioMask`) was always valid.

Aligned the clean audio and ground truth using cross-correlation, resulting in correct SNR output values

Challenges

Variable-length training sequences

RMS loudness imbalance

Model expecting fixed feature counts

Direct clean-speech prediction failed
(predicted the *clean magnitude spectrogram directly*)

Solutions

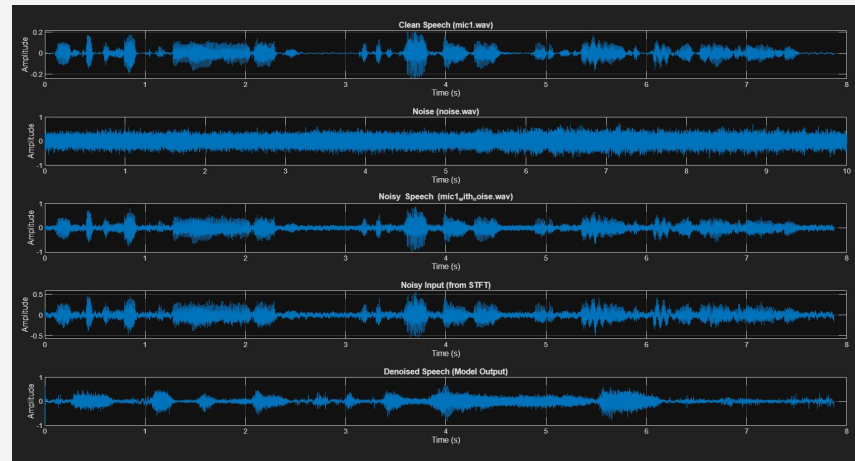
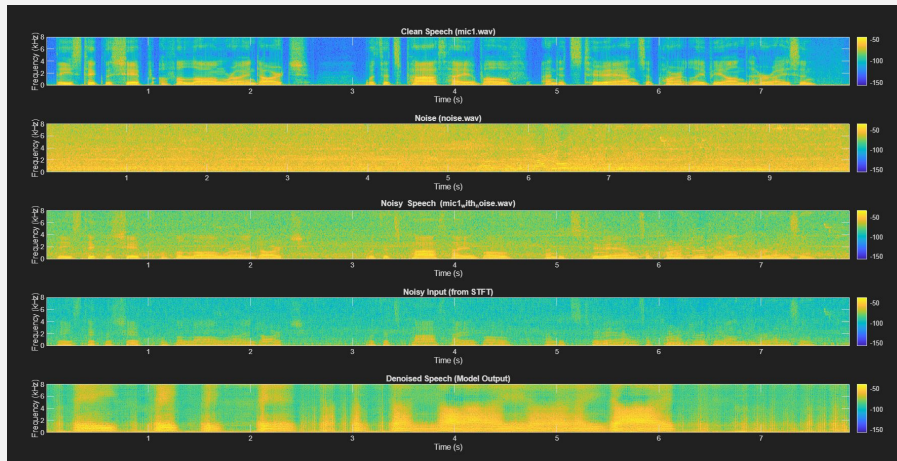
Switching to cell arrays for **XTrain** and **YTrain** allowed true variable-length sequence training for LSTMs.

We normalized the denoised signal using RMS scaling to ensure consistent loudness

We parameterized it by **numFeatures**, making the model flexible for any STFT configuration

Switched to predicting a ratio mask, dramatically improving reconstruction quality.

First Pass



(garbage output)

Challenges

- **Frame misalignment between predictors, phase, and model output**

- SOLUTION: We fixed this by enforcing a consistent frame count using

```
minFrames = min(size(mixtureMag,2),
size(ratioMask,2),
size(mixedPhase,2)).
```

- **Half-spectrum vs. full-spectrum reconstruction mismatches**

- SOLUTION: We rebuilt the missing negative-frequency bins using conjugate symmetry to ensure a valid complex STFT.

- **Magnitude and mask matrix size inconsistencies**

- SOLUTION: Clipping and resizing ensured that element-wise multiplication (`mixtureMag .* ratioMask`) was always valid.

- SOLUTION: **RMS loudness imbalance**
- SOLUTION: We normalized the denoised signal using RMS scaling to ensure consistent loudness.

- **Variable-length training sequences**

- SOLUTION: Switching to cell arrays for `XTrain` and `YTrain` allowed true variable-length sequence training for LSTMs.

- **Model expecting fixed feature counts**

- SOLUTION: We parameterized it by `numFeatures`, making the model flexible for any STFT configuration

- **Direct clean-speech prediction failed**

- Our first approach trained the model to predict the *clean magnitude spectrogram directly*.
- Although the plots looked visually “clean,” the reconstructed audio was unusable due to phase mismatch and magnitude instability.
- We switched to predicting a **ratio mask**, which keeps

Future Recommendations

- **Fix Data Distribution:**

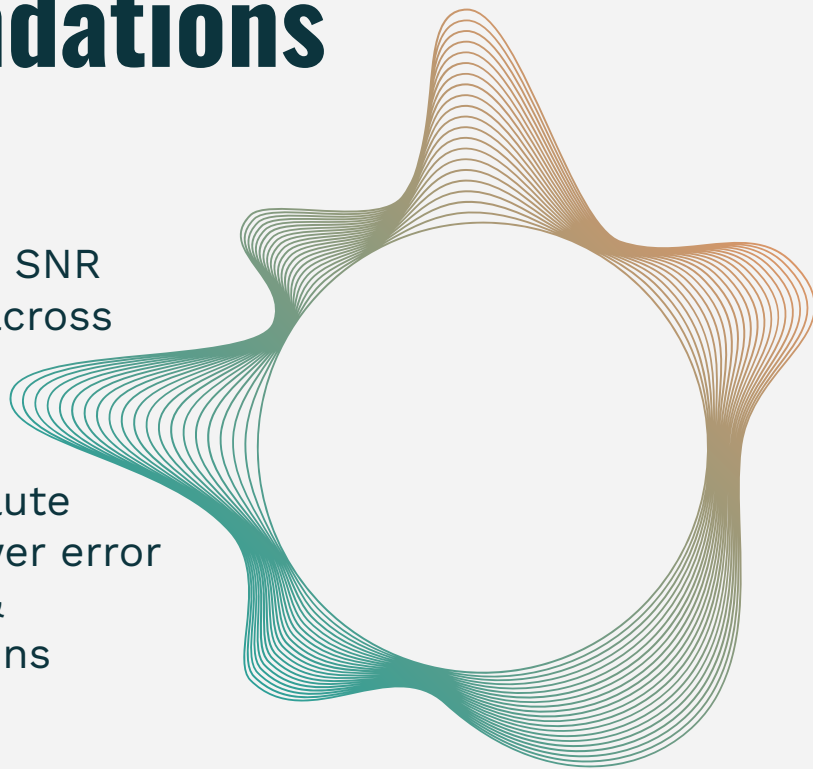
- Identify bug in mixing code
- Create new data with balanced SNR
- Retrain model and reevaluate across SNR levels

- **Address Amplitude Issue**

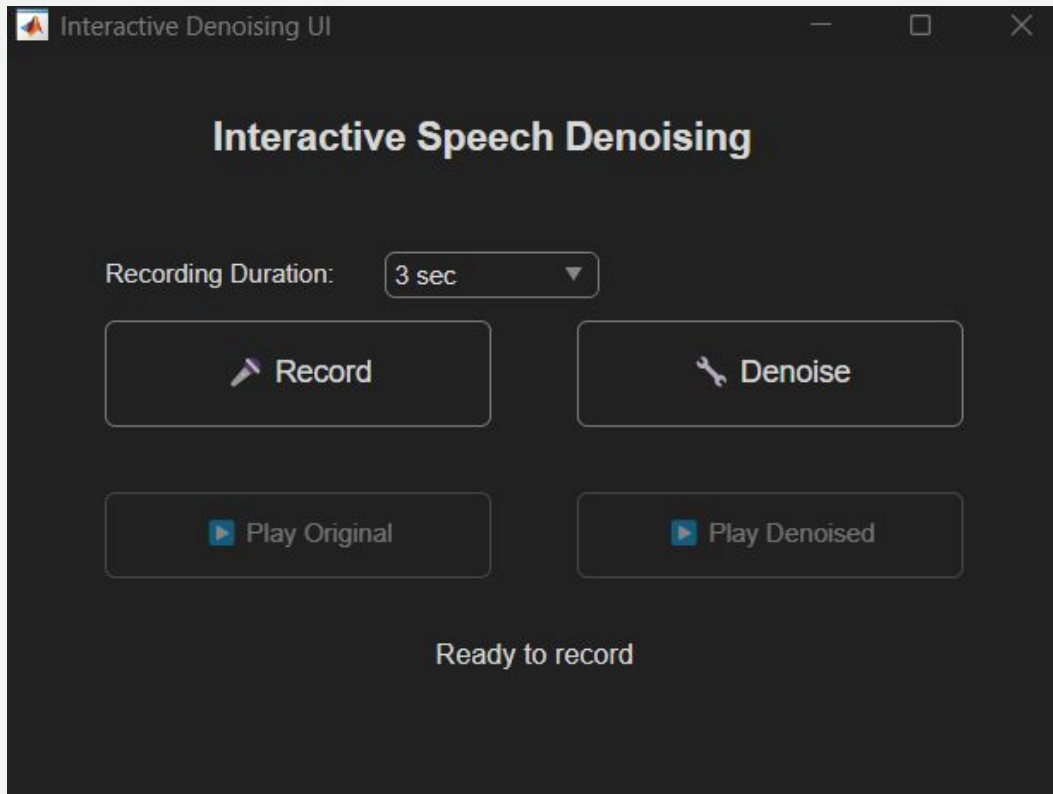
- Currently: MSE measures absolute magnitude: lower volume = lower error
- Experiment w/scale-invariant & perceptual training loss functions
 - SI-SNR, PESQ

- **Train on more diverse samples**

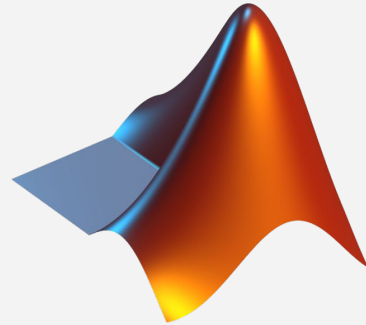
- Different languages, multiple speakers (cocktail-party separation)



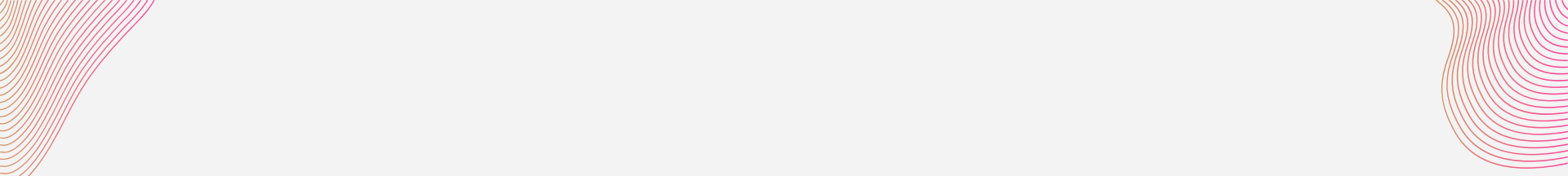
Interactive Speech Denoiser



Resources We Leveraged



Resources

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Thank You

<https://github.com/BTTAI-9/Team-9>

The bottom of the slide is decorated with a dense, horizontal row of thin, vertical grey lines of varying heights, creating a textured, bar-chart-like effect.